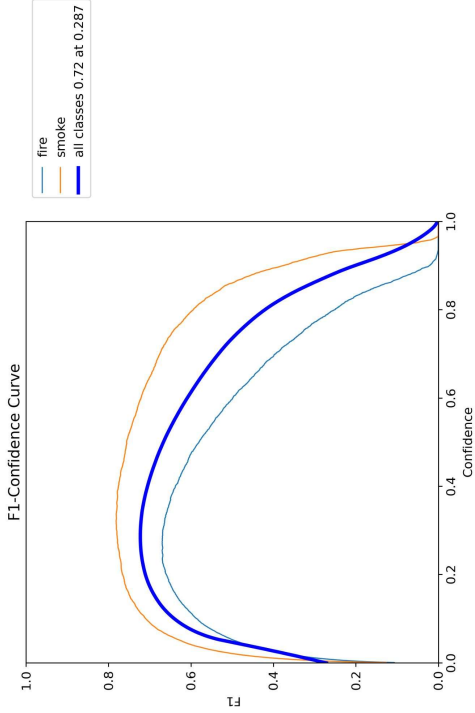
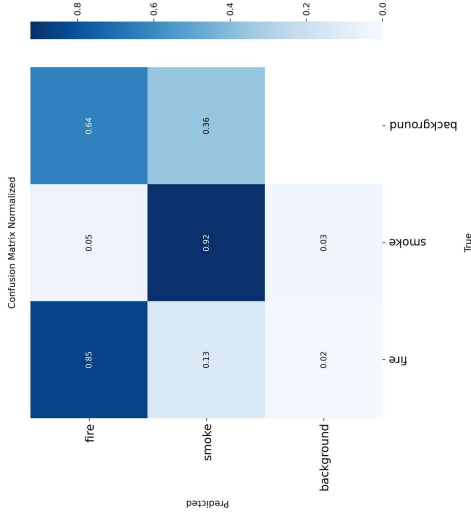
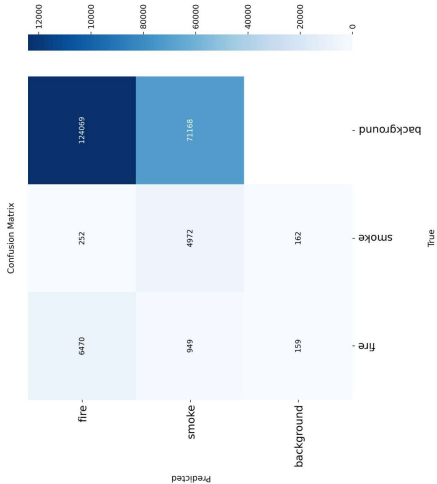


# Wesley Batista - Métricas

map50: 0.772, Precision(all): 0.786, Recall(all): 0.671

Dataset: total:95k, fire(only):12.5k, smoke(only):23.3k, fire+smoke:20.1k, null:39.1k

train(70%), val(15%), test(15%)



## Confusion Matrix

## Normalized Confusion Matrix

## F1 Curve

best: 0.72 F1 at

confidence 0.287

$\text{PRECISION} = \text{TP} / (\text{TP} + \text{FP})$   
(background, other classes)

$\text{RECALL} = \text{TP} / (\text{TP} + \text{FN})$   
(missed class)

Imagens do Dataset



Figure 5. Sample images from FASDD.

Guideline das Anotações

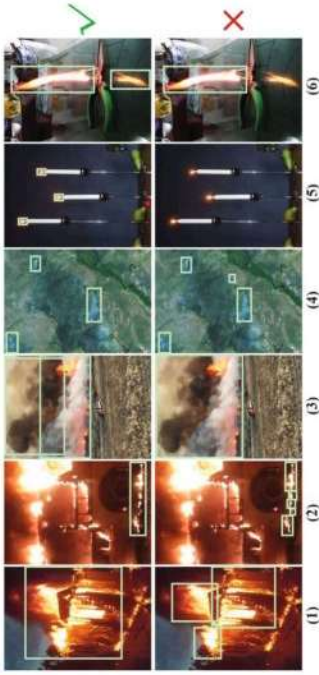


Figure 3. Annotation guidelines for the FASDD.

Benchmarks

Model: yolo26n

Parameters: epochs=100, imgsz=640, batch=256, workers=12, patience=30

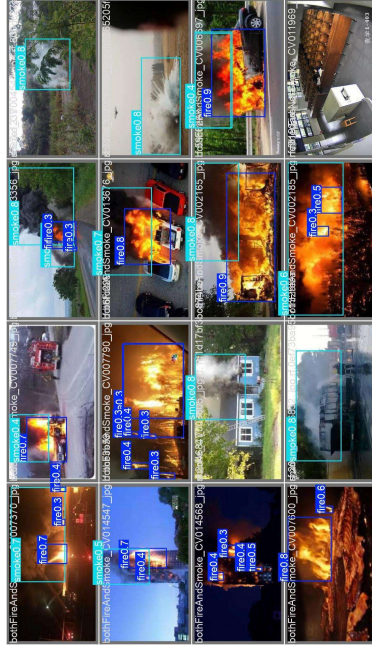
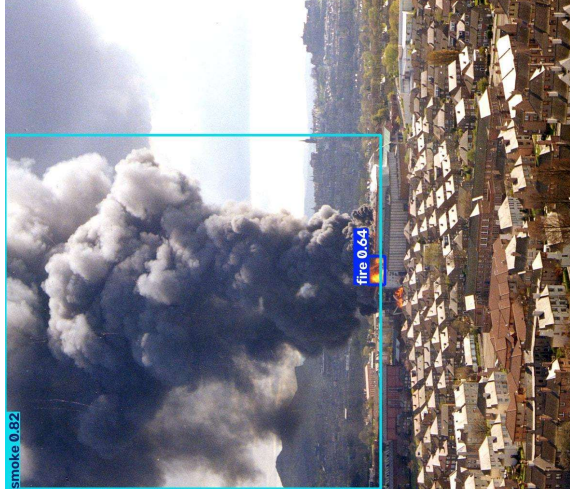
map50(all):0.772, map50(fire):0.719, map50(smoke):0.825

precision(all):0.786, precision(fire):0.733, precision(smoke):0.839

recall(all):0.671, recall(fire):0.615, recall(smoke):0.727

| Table A1. Accuracy evaluation results on FASDD for models with different basic architectures. |            |       |                               |                  |                |            |               |                |           |            |               |                |      |
|-----------------------------------------------------------------------------------------------|------------|-------|-------------------------------|------------------|----------------|------------|---------------|----------------|-----------|------------|---------------|----------------|------|
| Dataset                                                                                       | Batch size | Epoch | Network architecture paradigm | Model            | Validation (%) |            |               |                | Test (%)  |            |               |                |      |
|                                                                                               |            |       |                               |                  | AP (fire)      | AP (smoke) | Recall (fire) | Recall (smoke) | AP (fire) | AP (smoke) | Recall (fire) | Recall (smoke) |      |
| FASDD                                                                                         | 8          | 36    | CNN                           | YOLOv5x          | 73.8           | 77.2       | 75.5          | 75.0           | 84.4      | 83.1       | 83.7          | 83.8           | 66.2 |
|                                                                                               |            |       |                               | InternImage      | 74.0           | 82.2       | 78.1          | 90.7           | 90.8      | 85.4       | 84.0          | 93.6           | 92.7 |
|                                                                                               |            |       | Transformer                   | DETR             | 68.8           | 80.3       | 74.5          | 89.9           | 96.2      | 83.8       | 80.4          | 92.7           | 97.2 |
|                                                                                               |            |       |                               | Swin Transformer | 72.8           | 83.4       | 78.1          | 91.5           | 92.8      | 86.3       | 84.2          | 94.1           | 94.1 |

# Wesley Batista - Detecções - Imagens



# Wesley Batista - Detecções - Vídeos

